

Hao Chen

CONTACT INFORMATION

West Lafayette, IN, USA

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EDUCATION

Purdue University, West Lafayette, IN

Ph.D. in Chemical Engineering

2022 - Present

- Research advisor: Dr. [Can Li](#)

- Research expertise: Machine Learning and Mathematical Optimization

University of Cambridge, Cambridge, UK

M.Phil. in Chemical Engineering and Biotechnology

2020 - 2021

University of Nottingham, Nottingham, UK

B.Eng. in Chemical Engineering

2016 - 2020

APPOINTMENTS

Amazon, Bellevue, WA

2025

Applied Scientist Intern, Last Mile Planning

Supervisor: Dr. Jin Ye

PUBLICATIONS

- [1] **H Chen**, C Qian, C Morris, A Lodi, C Li. Solving Max-Cut to Global Optimality via Feasibility-Preserving Graph Neural Networks. *arXiv* (2026).
- [2] A Ramanujam, A Elyoumi, H Chen, SM Kompalli, AS Ahluwalia, S Pal, DJ Papageorgiou, C Li. SafeOR-Gym: A Benchmark Suite for Safe Reinforcement Learning Algorithms on Practical Operations Research Problems. *Transactions on Machine Learning Research* (2026).
- [3] H Kim, H Chen, C Li, JM Lee. TalkToAgent: A multi-agent LLM Framework for natural language explanation of reinforcement learning policies. *Computers & Chemical Engineering*. (2026).
- [4] GE Constante-Flores, **H Chen**, C Li. Enforcing Hard Linear Constraints in Deep Learning Models with Decision Rules. *The Thirty-ninth Annual Conference on Neural Information Processing Systems* (2025).
- [5] **H Chen**, GE Constante-Flores, KSI Mantri, SM Kompalli, AS Ahluwalia, C Li. OptiChat: Bridging Optimization Models and Practitioners with Large Language Models. *INFORMS Journal on Data Science*. (2025).
- [6] A Khan, R Nahar, H Chen, GE Constante-Flores, C Li. FaultExplainer: Leveraging Large Language Models for Interpretable Fault Detection and Diagnosis. *Computers & Chemical Engineering*. (2024).
- [7] HH Chiu, A Varghese, K Shao, YC Lu, R Nahar, H Chen, Q Deng, X Bao, C Li. scChat: A large language model-powered co-pilot for contextualized single-cell RNA sequencing analysis. *AIChE Journal*. (2024).
- [8] **H Chen**, GE Constante-Flores, C Li. Physics-informed neural networks with hard linear equality constraints. *Computers & Chemical Engineering*. (2024).
- [9] **H Chen**, GE Constante-Flores, C Li. Diagnosing Infeasible Optimization Problems Using Large Language Models. *INFOR: Information Systems and Operational Research*. (2024).
- [10] Y Yan, WI Shin, H Chen, et al. A recent trend: application of graphene in catalysis. *Carbon Lett.* (2021).

CONFERENCE PRESENTATIONS	Chen, H. (2025). Deep Learning Models with Hard Linear Inequality Constraints. Paper presented at the 2025 <i>AIChE Annual Meeting</i> , Boston, MA (CAST Director's Student Presentation Award Finalist)	
	Chen, H. (2024). Self-supervised Learning for Constrained Optimization with Hard Linear Constraints. Paper presented at the 2024 <i>INFORMS Annual Meeting</i> , Seattle, WA	
	Chen, H. (2024). GPU Accelerated Approximation Algorithm for Multi-Parametric Linear Programming. Paper presented at the 2024 <i>AIChE Annual Meeting</i> , San Diego, CA	
	Chen, H. (2024). Physics-Informed Neural Networks with Hard Linear Equality Constraints. Paper presented at the 2024 <i>AIChE Annual Meeting</i> , San Diego, CA	
	Chen, H. (2024). Diagnosing Infeasible Optimization Problems Using Large Language Models. Paper presented at the 2024 <i>AIChE Annual Meeting</i> , San Diego, CA	
HONORS AND AWARDS	Bilsland Dissertation Fellowship	2026
	Purdue Graduate Student Government (PGSG) Travel Grant	2025
	Ryland Travel Grant, School of Chemical Engineering, Purdue University	2025
	Purdue Graduate Student Government (PGSG) Travel Grant	2024
	Chao Travel Grant, School of Chemical Engineering, Purdue University	2024
	Nottingham Engineering Excellence Scheme (Top 1.0%), University of Nottingham	2019
	British Petroleum Prize, University of Nottingham	2019
	Provost's Scholarship (Top 1.5%), University of Nottingham	2018
	Dean's Scholarship (Top 10%), University of Nottingham	2017
PROFESSIONAL ACTIVITIES	Reviewer for <i>Manufacturing & Service Operations Management</i>	2026
	Reviewer for <i>Computer & Chemical Engineering</i>	2025-2026
	Reviewer for <i>Digital Chemical Engineering</i>	2025-2026
	Reviewer for <i>Applied Energy</i>	2025
	Reviewer for <i>Journal of Aviation Technology and Engineering</i>	2025
	Reviewer for <i>ACS In Focus</i>	2024
	Reviewer for <i>Reviews in Chemical Engineering</i>	2024